

EVIDENCE LOCKER

Solution Synopsis / Executive Summary

A cyber-fraud complaint support platform for India

Prepared from the current product implementation

Evidence Locker helps victims and authorized support teams convert fragmented cyber-fraud information into a structured, reviewable complaint package. It combines guided complaint capture, evidence metadata, integrity hashing, custody records, role-based workflows, and exportable documentation in one system.

Document scope

This summary describes the problem, solution, operating model, security posture, technology architecture, expected value, and important boundaries of the platform. It is not legal advice and does not guarantee FIR registration, investigation outcomes, or conviction.

1. Executive Summary

Cyber-fraud victims often need to coordinate with banks, platforms, cyber cells, police stations, and support organizations while preserving information that may be distributed across phones, email, screenshots, payment records, and online accounts. The result can be incomplete narratives, weak evidence context, unclear ownership, and slow handoffs.

Evidence Locker addresses this coordination problem with a secure, role-aware workflow that turns an incident into a consistent complaint record and evidence packet. It keeps the victim's story, supporting files, machine-extracted metadata, cryptographic hashes, status changes, and handoff events connected to a single complaint.

2. The Proposed Solution

- Guided complaint creation for common categories including UPI fraud, phishing, social-media harassment, loan-app fraud, and other cyber incidents.
- Evidence intake that accepts common image, video, audio, document, text, archive, and spreadsheet formats within controlled upload limits.
- Automatic metadata extraction where available, together with manually entered context such as transaction identifiers, source device, platform, and collection limitations.
- SHA-256 hashing and custody events to make post-upload integrity and handling history visible to reviewers.
- Role-based collaboration for victims, support or NGO helpers, police officers, and administrators.
- Complaint package generation for review, forwarding, and submission preparation, including a structured summary, evidence vault, custody table, and Section 65B draft template.

3. Intended Outcomes

- Improve the completeness and consistency of cyber-fraud complaints.
- Reduce avoidable back-and-forth during support and police review.
- Preserve evidence context and make handling history auditable.
- Give authorized users a clear view of status, ownership, next actions, and handoffs.
- Provide a practical foundation for escalation to the 1930 cyber-fraud helpline, local cyber cells, and relevant national workflows.

4. Users and Operating Model

Victim	Creates a complaint, supplies incident facts, uploads evidence, and tracks progress.
Support / NGO Helper	Assists with complaint preparation, evidence context, and supporting documentation.
Police Officer	Reviews submissions and evidence, records decisions, and requests clarification or forwards cases.
Administrator	Oversees users, policies, audit records, and access control.

5. End-to-End Workflow

- 1 Create complaint**
Capture victim identity, location, incident date, fraud category, and category-specific facts.
- 2 Upload evidence**
Attach files and collect supporting descriptions, source details, and transaction context.
- 3 Extract and verify metadata**
Read available metadata, merge manual fields, and calculate a SHA-256 integrity hash.
- 4 Review and collaborate**
Use role-aware access, status changes, remarks, and clarification requests.
- 5 Record custody and handoff**
Log evidence actions and record forwarding details, receiving unit, officer, and remarks.
- 6 Generate package**
Export a police-ready complaint packet for review or submission preparation.

6. Technology Architecture

The platform is implemented as a web application with a React and Vite frontend, a Node.js and Express backend, and MongoDB persistence through Mongoose. A mobile-oriented frontend and Android wrapper are also present in the repository for future or parallel delivery channels.

Presentation	React, React Router, Vite, Tailwind CSS
Application services	Node.js, Express, REST-style route modules
Data layer	MongoDB with Mongoose models for users, complaints, evidence, sessions, OTP challenges, and audit logs
Evidence services	Multer memory uploads, GridFS storage, exifr metadata extraction, SHA-256 hashing
Communications	Nodemailer-based OTP delivery and chatbot assistance through the chat route
Document services	pdf-lib-based complaint package generation

7. Security and Trust Controls

- OTP-based authentication with JWT session tokens and role-based authorization.
- Encrypted storage of sensitive complaint contact fields and controlled decryption for authorized responses.
- Audit logging for authentication, complaint, evidence, and workflow actions.
- HTTP security headers including content-type sniffing protection, frame restrictions, and a restrictive referrer policy.
- File-type validation, upload count and size limits, and rejection of unsupported evidence types.
- Retention fields and cleanup support for closed complaints and stored evidence.

8. Expected Value

- For victims: a guided, less fragmented way to document what happened and preserve supporting material.
- For support teams: a shared workspace that reduces duplicate data entry and makes assistance traceable.
- For police reviewers: a consistent packet with incident facts, evidence references, integrity hashes, custody history, and clear handoff information.
- For administrators: auditable access and operational visibility across users, complaints, evidence, and status transitions.

9. Important Boundaries and Risks

- The platform improves readiness and documentation; it cannot guarantee FIR registration, police acceptance, investigation progress, recovery of funds, or conviction.
- A hash calculated after upload demonstrates integrity from the point of hashing and storage; it cannot, by itself, prove that a file was original before upload.
- Metadata availability depends on the source file and device. Missing metadata must remain visible rather than being treated as proof of absence.
- The Section 65B certificate included in generated packets is a draft template and requires lawful review, completion, and attestation before use.
- Production deployment requires careful secret management, HTTPS, database access controls, rate-limit tuning, backup policy, monitoring, and privacy governance.

10. Success Measures

- Percentage of complaints completed with all mandatory and fraud-specific fields.
- Percentage of uploaded evidence items with usable metadata and recorded hashes.
- Time from complaint creation to package-ready status.
- Reduction in clarification cycles caused by missing incident or evidence context.
- Audit completeness for authentication, evidence handling, and complaint handoffs.

11. Conclusion

Evidence Locker provides a practical digital foundation for cyber-fraud complaint preparation in India. Its central contribution is the connection of narrative, evidence, metadata, integrity, custody, collaboration, and document generation into one role-aware workflow.

By improving the quality and traceability of information before it reaches a reviewer, the platform can help victims and support organizations spend less time reconstructing events and more time taking informed next steps. Its value is strongest when used as a preparation and documentation layer alongside official police, cyber-cell, banking, and legal processes.

Positioning statement

A secure, auditable, and user-centered workspace for turning cyber-fraud incidents into structured complaint and evidence packages.

Disclaimer

This document is a product and solution summary based on the current repository implementation. It is not legal advice, a police acceptance document, or a guarantee of any legal or investigative outcome.

12. AI Chatbot Assistance

Evidence Locker includes an authenticated AI chatbot powered by Google Gemini. It helps users understand digital incidents, identify missing facts, organize timelines, prepare factual report drafts, and understand evidence-preservation steps.

Role in the solution

- Guides first-time reporters through calm, relevant follow-up questions.
- Helps convert unstructured experiences into factual timelines and report drafts.
- Can use authorized evidence metadata linked to an incident, including filenames, hashes, capture times, upload times, and de
- Receives metadata only through the chat route; underlying evidence files are not sent to the model by this feature.

Trust and boundaries

The chatbot is an information and documentation assistant, not a lawyer, police officer, investigator, forensic examiner, or emergency service. It must not fabricate facts, authenticate evidence, provide definitive legal conclusions, or guarantee FIR registration, investigation outcomes, fund recovery, or conviction.

Implementation and operational safeguards

The backend requires authentication, bounds conversation history, limits message size, supports a configurable Gemini model, applies request timeouts, and returns explicit errors when the service is unavailable or not configured. Users and authorized reviewers must verify AI output against the original facts and evidence.

Assessment

The AI chatbot strengthens the solution by improving accessibility and documentation quality. Its value is highest when it helps users ask better questions and identify missing information while keeping human and official review in control.